For Examiner's Use

7 **(a)** Explain what is meant by the phase difference of two waves of the same frequency.

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……

…… [1]

(b) An experiment to investigate two-source interference using light waves is set up as shown in Fig. 7.1. Three slits S_0, S_1 and S_2 of equal width are used with a lamp that is attached with a red filter.

Fig. 7.1 (not to scale)

(i) For observable interference, suggest why slit S_o is needed when the light source used is a lamp.

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…… [2]

(ii) Draw on Fig. 7.1 lines to represent the rays of light from S_o to the point C on the screen which is equidistant to S_1 and S_2. [1]

(iii) State the path difference of the light reaching point C.

path difference = …… [1]

For Examiner's Use

(iv) **1.** The amplitude of the light at point C on the screen in Fig. 7.1 is ***A***.

State and explain the new amplitude at point C on the screen when a piece of polariser is placed in front of S_2.

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..... [2]

2. Describe the changes to the interference pattern on the screen if the red filter in Fig. 7.1 is replaced with a blue filter.

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..... [2]

(c) **(i)** Explain the formation of a stationary wave.

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..... [2]

For Examiner's Use

(ii) A stationary wave is setup in a one end open tube using a speaker which is connected to a signal generator as shown in Fig. 7.2.

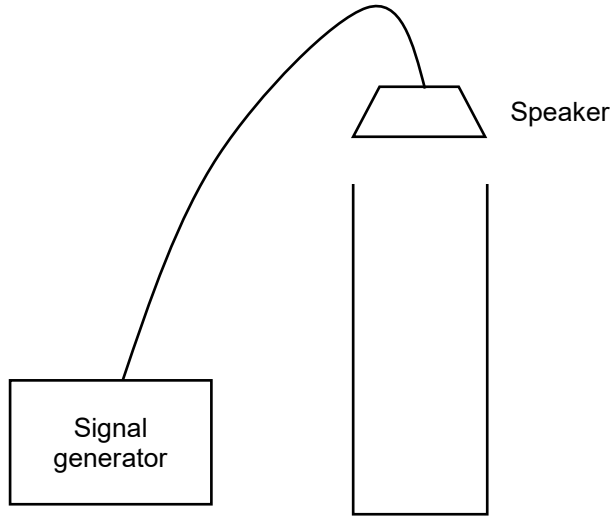

Fig. 7.2

The length of the tube is 90 cm long and the wavelength of the sound is 120 cm. A small microphone which senses pressure variation is placed in the tube to measure the frequency of the stationary wave formed.

The microphone is connected to a cathode-ray oscilloscope (c.r.o). When the microphone picks up the largest pressure variation, the waveform obtained on the c.r.o. is shown in Fig. 7.3.

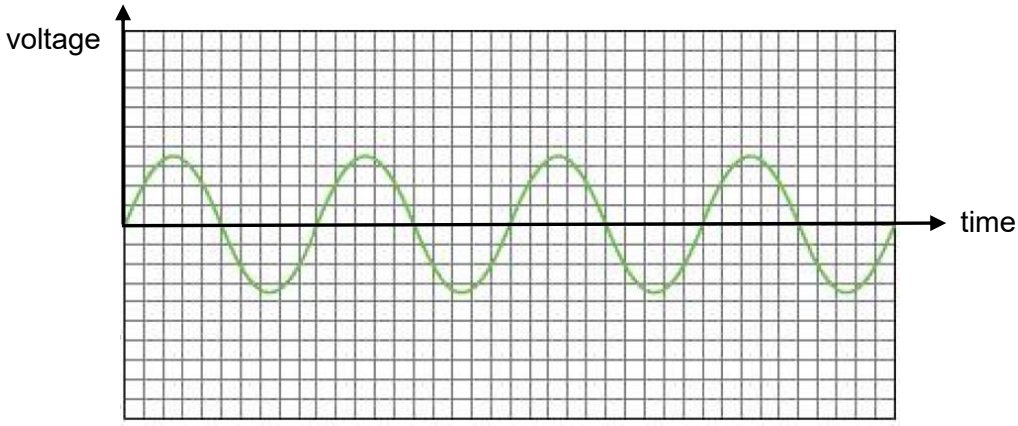

Fig. 7.3

For Examiner's Use

1. State and explain the distance in the tube from the open end for the microphone to be placed so that the waveform in Fig. 7.3 is obtained.

Include in your answer a diagram of the stationary wave setup in the air column of the tube. Label your diagram.

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..... [3]

2. The wavelength of the sound from the speaker is increased to 330 cm. State and explain if a stationary wave can be formed.

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..... [2]

For Examiner's Use

(d) A monochromatic laser is incident on a diffraction grating.

By drawing a suitable well-labelled diagram, describe how you would determine the wavelength of the laser light.

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…… [4]